Characterizing human semantic navigation as trajectories in embedding space

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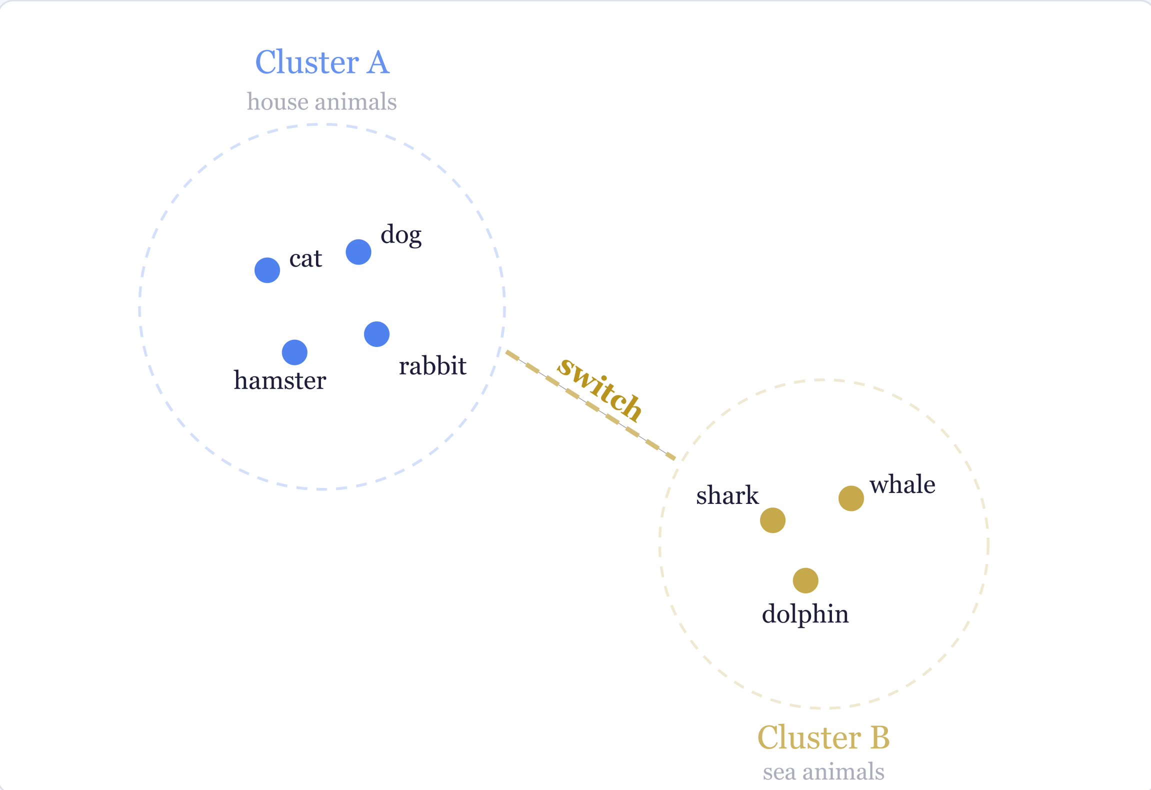
Semantic navigation: moving through concepts

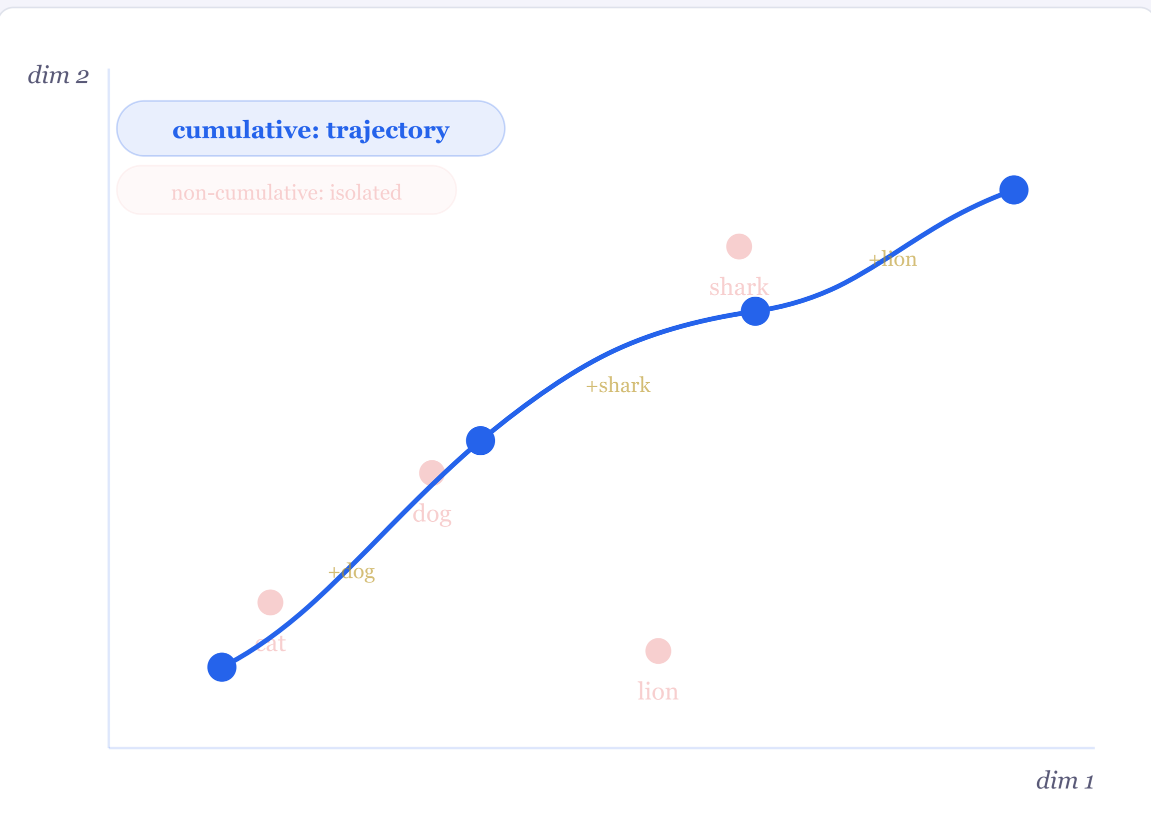


Semantic retrieval is often characterized as *clustering* and *switching*; traditional pipelines are labor-intensive, heterogeneous, and miss the step-by-step granularity of how meaning unfolds over time.

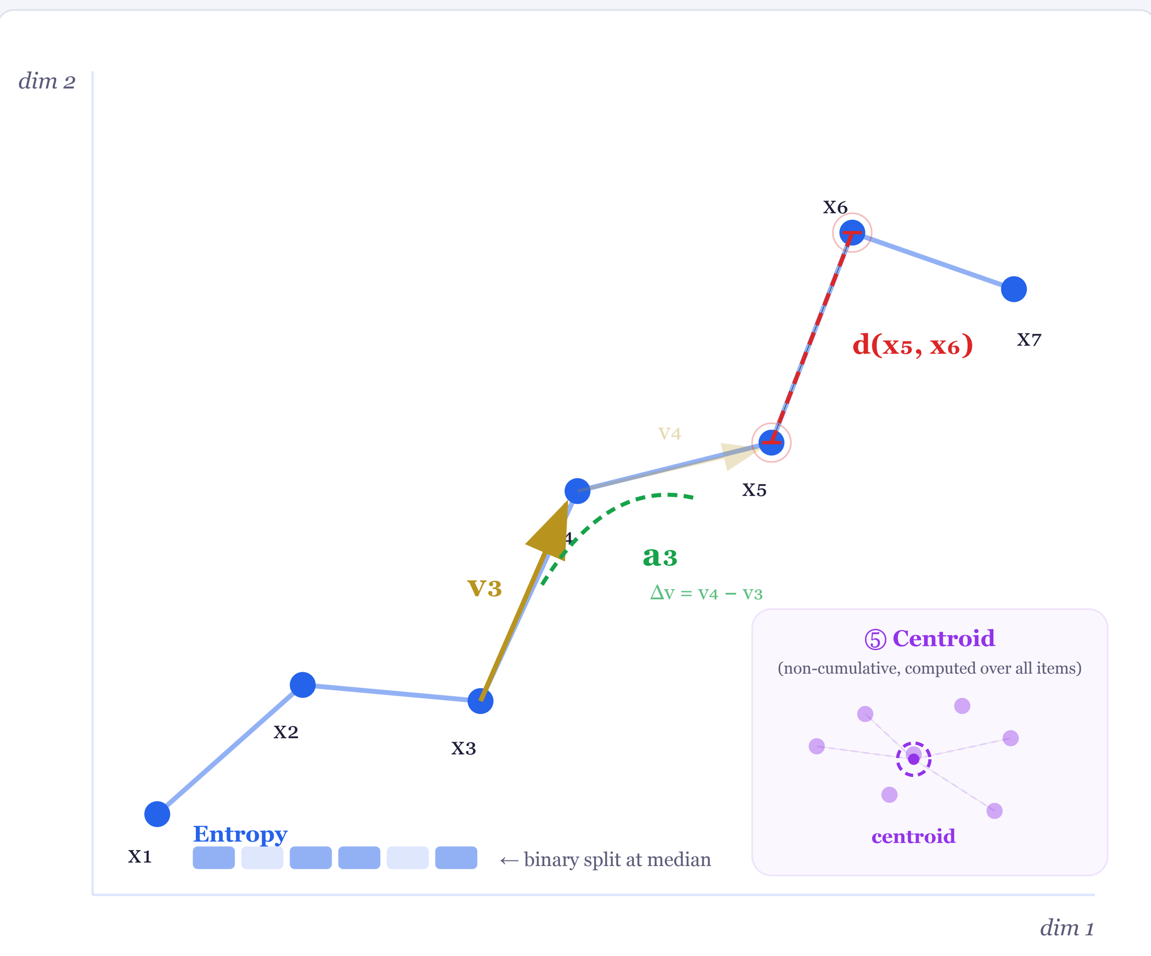
→ What if we framed semantic retrieval as a trajectory through geometric space?

Cumulative embeddings encode search history

Each step encodes the full prefix: embed `"cat"` → `"cat dog"` → `"cat dog shark"` ... capturing dependencies and working memory. Result: a trajectory per participant × concept pairs.



Five physics-inspired trajectory metrics



① Distance to next

Cosine distance between consecutive points. Semantic jump size.

② Velocity $v_t = x_{t+1} - x_t$

Direction + magnitude of each step.

③ Acceleration $a_t = v_{t+1} - v_t$

Low → stable cluster.
High → erratic switch.

④ Entropy

Shannon entropy of median-split steps. Predictability of the search.

⑤ Distance to centroid

Distance to mean position of all items. Dispersion of the search.

Four datasets, four languages, two tasks

Statistics: GLMMs with Tukey HSD post-hoc.

A Neurodegenerative
ES-CL · N=76 · Property Listing · HC / PD / bvFTD

B Swear Fluency
EN · N=274 · Verbal Fluency · Swear / Animals / Letters

C Cross-Linguistic PLT
IT (N=69) & DE (N=73) · 10 categories · Property Listing

Embedding models

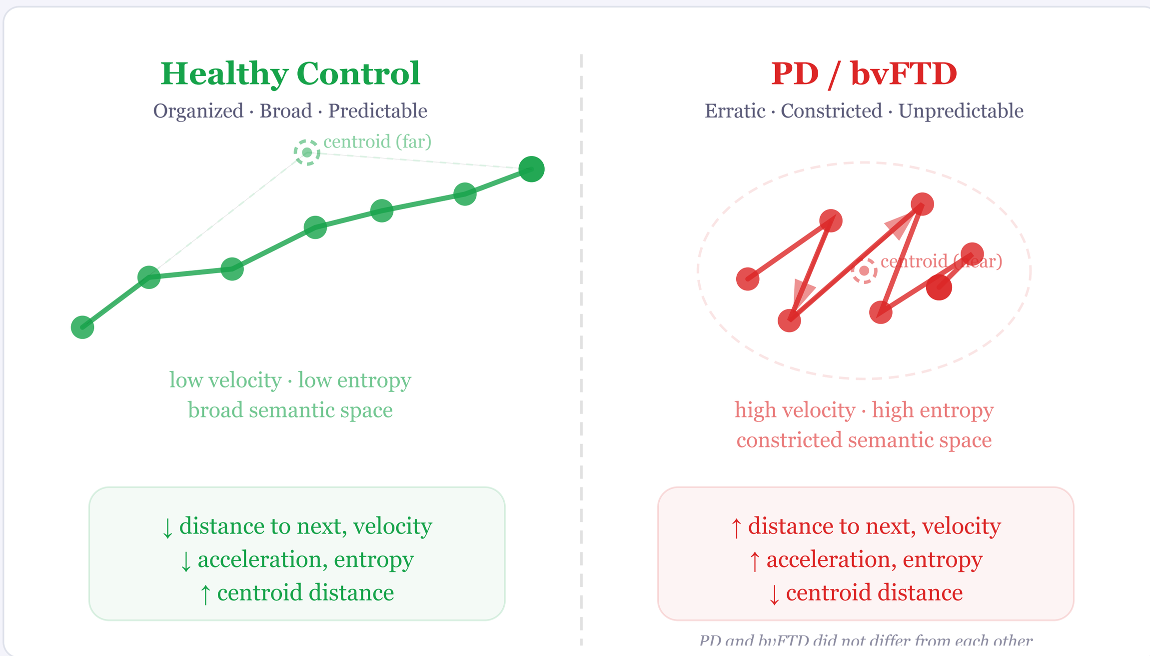
- OpenAI text-embedding-3-large
- Google text-embedding-004
- Qwen3-Embedding-0.6B
- fastText (baseline, non-cumul.)

All results reported with OpenAI.

Design choices

Causal and bidirectional attention
Cumulative vs. non-cumulative comparison
ZCA-whitening tested for anisotropy

Patient trajectories: erratic yet constricted



↑ Higher in patients

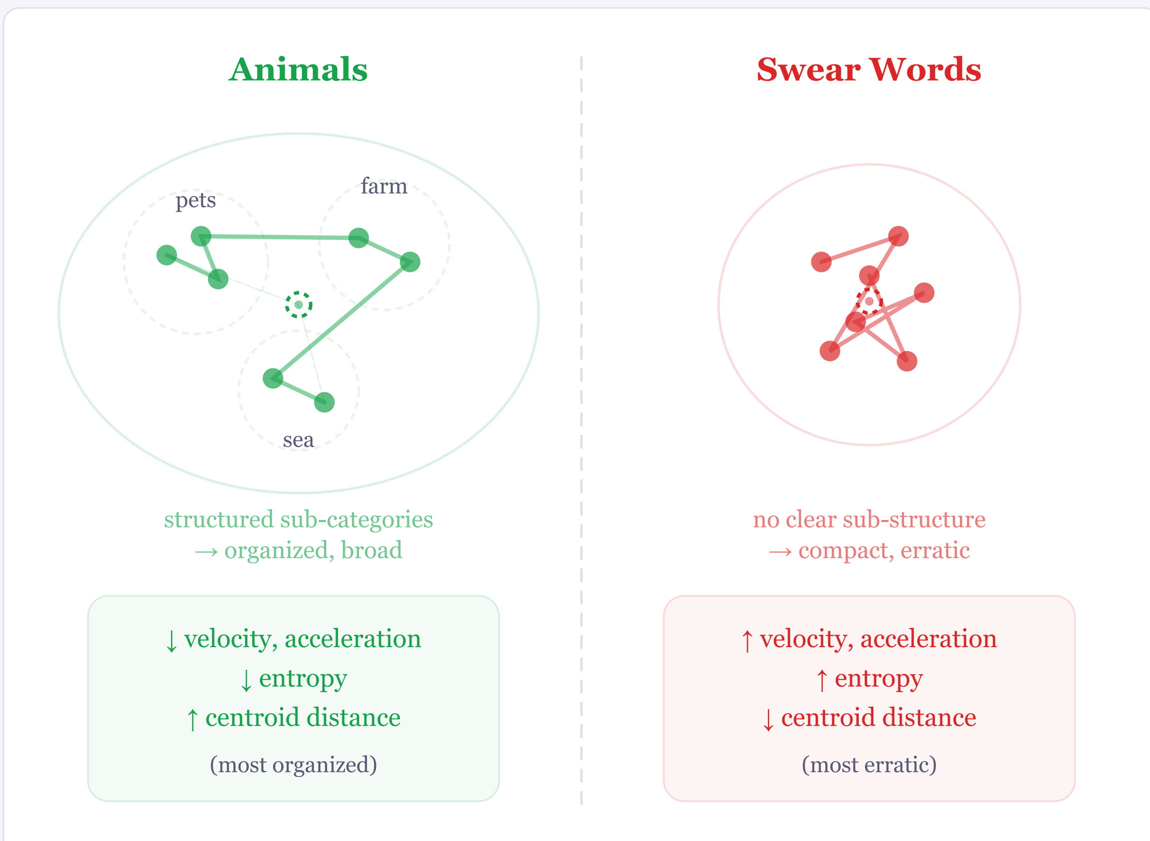
Dist. to next — larger semantic jumps
Velocity — erratic movement
Acc. — abrupt direction changes
Entropy — unpredictable search

↓ Lower in patients

Distance to centroid — search confined to a tighter neighborhood despite being more volatile

A kinematic signature of executive dysfunction — patients navigate semantic space with volatile, erratic steps yet remain confined to a narrow neighborhood.

Category structure shapes semantic kinematics



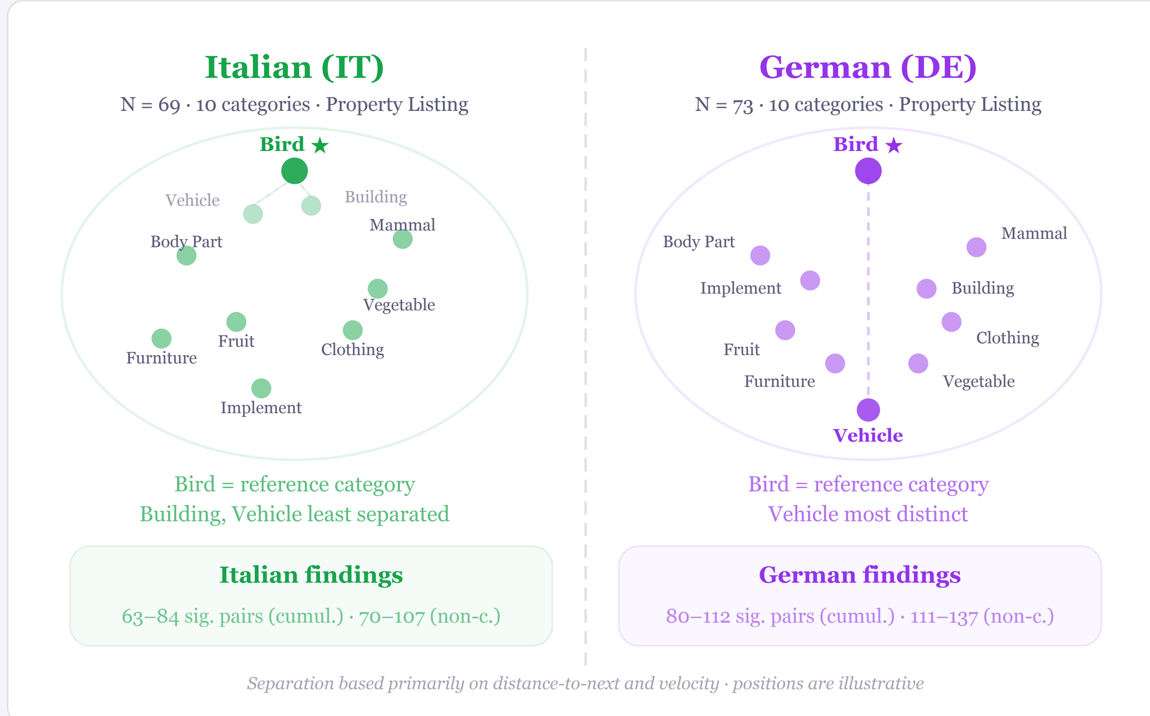
Animals

Lowest kinematic values, highest centroid distance. Structured sub-categories (pets, farm, sea...) allow **organized exploitation**.

Swear Words

Highest kinematic values across all metrics. Taboo lexicons lack sub-category structure → **high variability in a compact space**.

Language/culture shape semantic organization



Italian (IT) · N=69

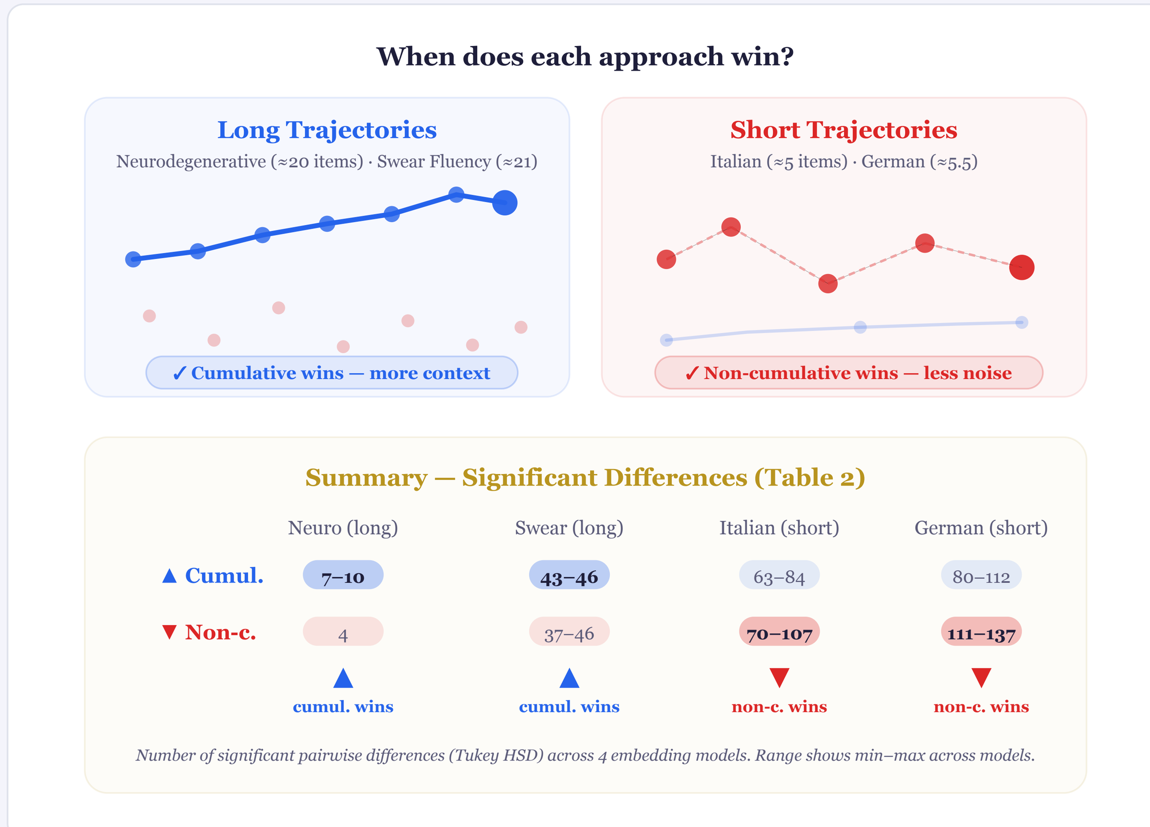
10 categories · Property Listing Task. Bird is reference. Building and Vehicle **least separated**. Category effects *selective*, not uniform.

German (DE) · N=73

Same 10 categories · Same protocol. Vehicle **most distinct** from Bird. Bird highest velocity & acceleration. Fruit vs. tool-like categories show strongest centroid gaps.

Same protocol, different effects; cultural/linguistic structure shapes organization.

Cumulative wins for long sequences



Cum. → long sequences

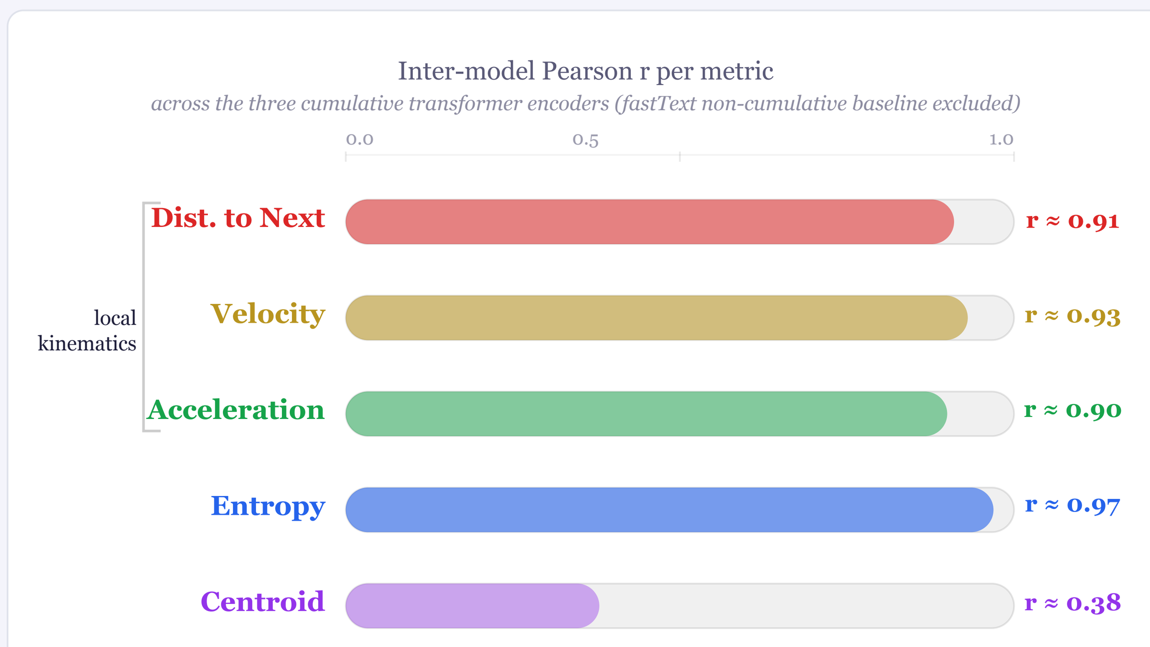
Longer trajectories provide rich context: **more significant group differences** with higher effect sizes.

Non-cum. → short sequences

With ~5 items, there is **little context** to accumulate. Point-to-point variation retains more discriminative signal.

Trajectory length determines which best exposes semantic structure.

Different models, convergent geometry



Towards a framework for semantic navigation

1. Cumulative metrics extend clustering/switching with fine-grained dynamics
2. Clinical utility — kinematic signatures of neurocognitive dysfunction
3. Discriminates categories & reveals language-specific organization
4. Model-robust — local kinematics converge across embedding architectures

Future: temporal timestamps · more mathematically robust metrics

Paper

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Code